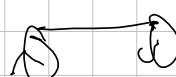
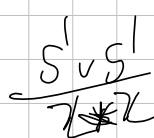
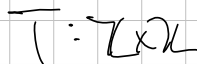
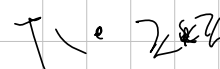
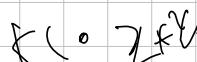


$(\mathbb{Z} \times \mathbb{Z})$   
 $aba^{-1}b^{-1} = e$   
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$   






[illegible]

$R/P^2$ : 0-cell:  $\{v\}$   
 1-cell:  $\{w\}$   
 2-cell:  $\{w\}$

$R/P^2$ :  $\mathbb{Z}_2$   $\langle a \mid a^2 \rangle$   
 $a^2 = 1$

$R/P^2$ : 0-cell:  $\{v\}$   
 1-cell:  $\{w\}$   
 2-cell:  $\{w\}$

$\pi_1(R/P^2) = \mathbb{Z}_2$